Smart Pet Feeder V2 - Dual MCU · Sheet 1/1 · Rev V2.0

KiCad-style schematic. Blue = UART bridge, orange = isolated servo power domain, single star ground at NT1. Scroll sideways or pinch to zoom.

Zoom in | Zoom out | Fit width

POWER / ISOLATION RULES (read before wiring)

A +5V_LOGIC (adapter ≥5 V/3 A) → U1 VIN, U2 5V, U7 VCC. Never feed it from BT1 or U10.
B +3V3 (U1 LDO) → U4, U5, U6, U8, R1/R2. HX711 + IR OUT stay ≤3.3 V (ESP32 GPIO is not 5 V tolerant).
C +VSERVO = BT1 (2S LiPo) → F1 → SW1 → U10 buck (5.5–6.0 V) → U9 only. MG996R window is 4.8–7.2 V:
1S LiPo (3.0–4.2 V) is too low, 2S fully charged (8.4 V) is too high → buck required.
D GND_LOGIC and GND_PWR meet at ONE star point (NT1). Servo return current never crosses logic ground.
E UART2 (U1) ↔ UART1 (U2): TX→RX, RX→TX, shared GND. Both 3.3 V logic, no level shifter.
F Servo wires: twisted V+/GND pair ≤30 cm, away from HX711 / load-cell cable (cross at 90°).
G U2 needs 5 V ≥2 A with a local 470 µF cap; U2 IO0 must float in run mode (flash jumper removed).

ARCHITECTURE V2.0

U1 WROOM: RTC, OLED, HX711, servo, IR, HC-SR04, state machine

U2 CAM: Wi-Fi, web UI, camera, TinyML

Link: UART 115200 8N1

U4 DS3231 RTC
VCC, SDA, SCL, GND, VBAT — I²C 0x68
BT2 CR2032
ZS-042 module: remove charge diode for CR2032 (non-rech.)

U5 OLED SSD1306
VCC, SDA, SCL, GND — I²C 0x3C

U6 HX711
RED E+, BLK E−, WHT A+, GRN A−, VCC, DT, SCK, GND — 24-bit ADC
LC1 Load Cell: E+, E−, A+, A−
Bowl rests on LC1 only (no frame contact)

U7 HC-SR04
VCC 5V, TRIG, ECHO, GND — hopper level
R3 1k, R4 2k
ECHO divider: 5 V → 3.3 V

U8 IR Obstacle
VCC, OUT, GND — SSR1017

R1 4.7k, R2 4.7k
I²C 400k

U1 ESP32-WROOM-32
3V3, VIN 5V, IO21 SDA, IO22 SCL, IO17 TX2, IO16 RX2, IO25 HX_DT, IO18 SERVO, IO26 HX_SCK, IO27 TRIG, IO34 ECHO, IO33 IR_OUT, GND, GND
Sensor Hub · real-time I/O
UART0 (USB) = debug/flash only
Strapping IO0/2/5/12/15 unused

UART BRIDGE
115200 8N1 · 3.3 V logic
TX→RX and RX→TX, common GND
SERVO PWM
GND — common reference
R5 1k, R6 1k

U2 ESP32-CAM
5V, IO14 TX1, IO13 RX1, GND
U0T GPIO1, U0R GPIO3, IO0
J3:TX, J3:RX, J3:IO0→GND
Wi-Fi AP/STA · REST · SSE
OV2640 + 4 MB PSRAM
TinyML (TFLite Micro)
IO16 = PSRAM CS: unused
IO13/14 = SD pins: no SD
Vision / IoT Hub
J3 = FTDI header, flash only; IO0→GND while uploading, then remove

U9 MG996R
SIG (OR), V+ (RD), GND (BR) — gate servo
+VSERVO, GND_PWR
C3 2200µF + C4 100nF at servo pins

DOMAIN A · LOGIC 5 V (never from LiPo/servo buck)
J1 5V ≥3A: +5V, GND — adapter / DC jack
C5 1000µF, C6 470µF at U2 5V pin
GND_LOGIC

DOMAIN C · SERVO (isolated)
BT1 2S LiPo: +, − — 7.4 V
F1 3A, SW1
U10 Buck 5.5–6.0V: IN+, IN−, OUT+, OUT− — ≥3A low-dropout
GND_PWR rail

star point NT1
servo return current stays on this rail
NT1 = ONLY tie between grounds

WIRE LEGEND
signal / +3V3 / +5V
UART bridge U1 ↔ U2
servo domain (+VSERVO / GND_PWR)
Domain B: +3V3 from U1 LDO → U4, U5, U6, U8, R1, R2

Title:	Smart Pet Feeder V2 - Dual MCU
File: TL_CIRCUIT_V2_0.kicad_sch	Sheet: 1/1
Company: Tinkering Lab TL'26 · Size A3	Date: 2026-10-13 · Rev: V2.0